Go

Language Basics

- Introduction to Go
  - Why use Go
  - History of Go
  - Setting up the Environment
  - Hello World in Go
  - `go` command
- Variables & Constants
  - var vs :=
  - Zero Values
  - const and iota
  - Scope and Shadowing
- Data Types
  - Numeric Types
    - Integers (Signed, Unsigned)
    - Floating Points
    - Complex Numbers
  - Boolean
  - Runes
  - Strings
    - Raw String Literals
    - Interpreted String Literals
  - Type Conversion
- Commands & Docs

Composite Types

- Arrays
- Slices
  - Capacity and Growth
  - make()
  - Slice to Array Conversion
  - Array to Slice Conversion
- Strings
- Maps
  - Comma-Ok Idiom
- Structs
  - Struct Tags & JSON
  - Embedding Structs

Conditionals

- Loops
  - for loop
  - for range
  - Iterating Maps
  - Iterating Strings
  - break
  - continue
  - goto (discouraged)
- Conditionals
  - if
  - if-else
  - switch

Functions

- Functions Basics
- Variadic Functions
- Multiple Return Values
- Anonymous Functions
- Closures
- Named Return Values
- Call by Value

Pointers

- Pointers Basics
- Pointers with Structs
- With Maps & Slices

Methods and Interfaces

- Methods vs Functions
- Pointer Receivers
- Value Receivers
- Interfaces
  - Interfaces Basics
  - Empty Interfaces
  - Embedding Interfaces
  - Type Assertions
  - Type Switch
- Generics
  - Why Generics?
  - Generic Functions
  - Generic Types / Interfaces
  - Type Constraints
  - Type Inference

Error Handling

- Error Handling Basics
- `error` interface
- errors.New
- fmt.Errorf
- Wrapping/Unwrapping Errors
- Sentinel Errors
- `panic` and `recover`
- Stack Traces & Debugging

Code Organization

- Modules & Dependencies
  - go mod init
  - go mod tidy
  - go mod vendor
- Packages
- Package Import Rules
- Using 3rd Party Packages
- Publishing Modules

Concurrency

- Goroutines
- Channels
  - Select Statement
  - Buffered vs Unbuffered
- Worker Pools
- `sync` Package
  - Mutexes
  - WaitGroups
- `context` Package
  - Deadlines & Cancellations
  - Common Usecases
- Concurrency Patterns
  - fan-in
  - fan-out
  - pipeline

Testing & Benchmarking

- `testing` package basics
- Table-driven Tests
- Mocks and Stubs
- `httptest` for HTTP Tests
- Benchmarks
- Coverage

Standard Library

- Race Detection
- I/O & File Handling
  - flag
  - time
  - encoding/json
  - os
  - bufio
  - slog
  - regexp
- go:embed for embedding

Ecosystem & Popular Libraries

- Building CLIs
  - Cobra
  - urfave/cli
  - bubbletea
- Web Development
  - net/http (standard)
  - Frameworks (Optional)
    - gin
    - echo
    - fiber
    - beego
- gRPC & Protocol Buffers
- ORMs & DB Access
  - pgx
  - GORM
- Logging
  - Zerolog
  - Zap
- Realtime Communication
  - Melody
  - Centrifugo

Go Toolchain and Tools

- Core Go Commands
  - go run
  - go build
  - go install
  - go fmt
  - go mod
  - go test
  - go clean
  - go doc
  - go version
- Code Quality and Analysis
  - go vet
  - goimports
  - Linters
    - revive
    - staticcheck
    - golangci-lint
  - Security
    - govulncheck
- Code Generation / Build Tags
  - go generate
  - Build Tags
- Performance and Debugging
  - pprof
  - trace
  - Race Detector
- Deployment & Tooling
  - Building Executables
  - Cross-compilation

Advanced Topics

- Memory Mgmt. in Depth
- Escape Analysis
- Reflection
- Unsafe Package
- Build Constraints & Tags
- CGO Basics
- Compiler & Linker Flags
- Plugins & Dynamic Loading